STAT 414 Final Project

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Abstract

This report uses various survey-based statistical methods to analyze trends in statistics and data science programs in US colleges and universities. We have five chosen research questions that aim to identify and compare the availability, characteristics, and requirements of statistics and data science programs among the different institutional strata. The goal of this research is to gain insight on information that will be useful to further developing educational programs within this emerging field.

Key Words: *statistics, data science, statistical methods, US colleges and universities*

Introduction

Statistics is defined by Brittanica as the science of collecting, analyzing, presenting, and interpreting data. With such a broad definition, it is understandable that the discipline of statistics is used in numerous fields and professions across the world. As the field of statistics (and similarly data science) continues to grow, there are becoming more and more available jobs for statisticians and data scientists. In fact, the fields of statistics and data analytics are among the fastest growing in the country. According to the U.S. Bureau of Labor Statistics, data scientists were projected to be the fourth fastest growing occupation in the United States at a 36% increase in jobs available (*Fastest Growing*, 2024). Even though statisticians were behind that number at an 11% increase, this is still far above the national average of 4.2% growth.

With such quick growth, it is important to analyze the institutions that offer statistics and data analytics degrees and courses. As statistics and data science (SDS) students, we are interested in finding out more about the prevalence of these types of classes in universities across the United States. We compiled a list of questions we were interested in, as we thought these would help give us an insight as to what kinds of universities offer SDS programs. There were four questions in particular that were asked at the university level:

1. How many universities and colleges offer a statistics major, and how does this vary across strata?
2. Do MSI schools offer data science or statistics major/minor at a different frequency than non-MSI schools?
3. Do selective schools have higher rates of data science major compared to inclusive schools?

4. Do selective schools offer both major and minor in statistics at a different frequency compared to inclusive schools?

One question was asked at the individual course level as well:

1. Is there a difference in the proportion of schools that require math prerequisites for statistics degrees and data science degrees?

Methodology

This exact analysis was the point of study for the Motivational Attitudes in Statistics and Data Science Education Research (MASDER) team. The team was composed of seven professors at Universities across the United States who received a grant to "investigate trends and attitudes in statistics and data science education." Using their grant money, they used the US Department of Education College Scorecard Database to take their sample on the population of interest (US colleges and universities). In the end, the MASDER group decided to take a stratified sample of the universities in the database. The eight strata each institution was sorted into was based on: whether the institution was minority-serving, the selectivity of the school, and the highest degree offered. The population in each strata was recorded (Nh), as well as the overall population of interest (N). Within each of the sampled universities, the group collected as much data as they could, including but not limited to:

- University name and UnitID in the database
- Whether a statistics major or minor was available
- Whether a data science major or minor was available
- Whether there were introductory statistics/data science courses offered

This data was then compiled into an excel file for the "university level" of data.

The researchers were not only interested in statistics at a university level though. From all of the sampled institutions, they found every statistics and data science course available in the course catalogs and compiled them into another dataset called courses. This excel file had data on:

- Course name and department
- Course classification
- Prerequisites for each class
- Whether a lab was offered or not

Using the datasets collected by MASDER, we were able to run analyses on the questions we had asked above. Though it was necessary to clean and mutate the data to investigate our questions, we are very grateful to the MASDER team for their hard work in creating the files in the first place.

The data was mainly clean and there was not much of a need for data cleaning. However, for simplicity reasons we merged University and Course dataset into one csv file through R. This

was done by removing the missing and duplications of the UNITID column in both datasets and eventually merging them to a single dataset by UNITID column.

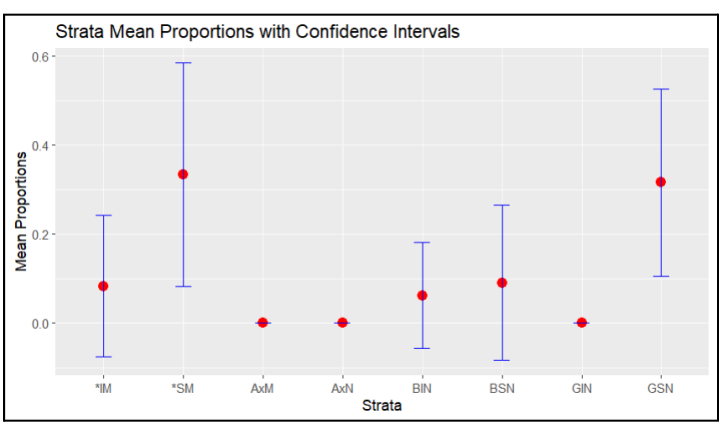
Results

1. **University Level**
   1.1. **How many universities and colleges offer a statistics major, and how does this vary across strata?**

Despite being an established science, statistics is an ever growing field. Naturally, we were curious as to how prevalent statistics majors are in universities across the country. In order to analyze this question, we had to turn our data into binary data. The new binary column we created would have a "0" if the university did not offer a statistics major and would have a "1" if it did. We were able to use R and its survey package to estimate the true proportion of each strata that offers a statistics major, as well as the estimated total number of schools that offer a statistics major. Below is a table of our results, grouped by strata:

Strata	Mean	SE
*IM	0.0833	0.0809
*SM	0.3333	0.1281
AxM	0	0
AxN	0	0
BIN	0.0625	0.0609
BSN	0.0909	0.0887
GIN	0	0
GSN	0.3158	0.1072

After getting our output, we created a graph of each mean and its corresponding 95% percent confidence interval using the ggplot package.



As we can see, the *SM and GSN strata have the highest estimated proportion at just above 30%. However, due to the standard errors of the data, we cannot say with accuracy which stratum has the highest proportion. This data still gives us an important insight into which types of universities are more likely to offer statistics majors and which ones are less likely to.

Another part of this question we were interested in analyzing was the total number of institutions that offer a statistics major. Using the svytotal and confint functions, R gave us a mean of 217.68 universities and a 95% confidence interval of (104.37, 330.99). This means we are 95% confident that the true number of colleges and universities that offer a statistics major is between 104 and 331.

Finally, we wanted to test if there was a statistically significant difference between proportion of *IM and *SM schools offering stats majors. Upon initial glance, it seems there is considerable overlap between the two confidence intervals, so we would not expect there to be a difference. We ran a t-test anyway to confirm or deny our suspicions. Using the mean and standard error values from above, along with code that ran a t-test, we found a p-value of 0.74. At a standard significance level of 95%, we would find a meaningful difference if the p-value was 0.05 or lower. Given our p-value here is much higher, we would conclude that there is not enough evidence to say that there is a significant difference between the proportion of *IM and *SM schools that offer statistics majors.

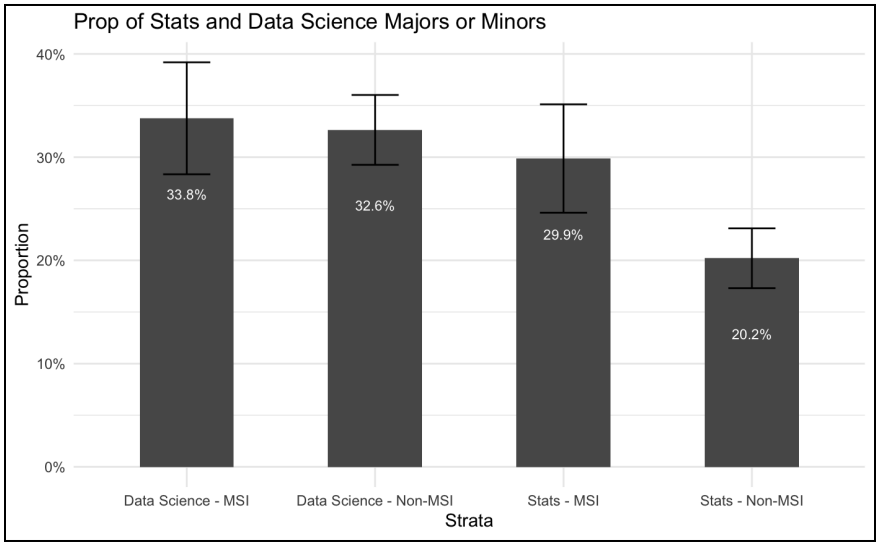
1.2. Do MSI schools offer data science or statistics majors/minors at a different frequency than non-MSI schools?

The first thing needed to answer this question is the proportion of schools that offer data science and statistics majors/minors. For this question, an assumption will be made that if a school offers either a major OR minor, it will be included in the proportion. To go about finding these numbers, we filtered all eight strata into two groups: MSI and non-MSI. We then created two binary columns, each representing whether or not a given school offered a statistics and data science program. We created the design within the MSI group using the svydesign function, and then the svymean to find the proportions of statistics and data science programs offered. This was then repeated in the non-MSI group and we found the corresponding proportions.

Program & Strata	Mean	SE	Confidence Interval
Statistics, MSI	0.2987	0.0525	(0.1958, 0.4016)
Data Science, MSI	0.3377	0.0542	(0.2313, 0.4440)
Statistics, non-MSI	0.2021	0.0290	(0.1453, 0.2589)

Data Science, non-MSI	0.3264	0.0338	(0.2601, 0.3928)

We also created a visualization of these proportions with the standard error to easily compare.



The last thing we did was run a t-test to determine if there is a significant difference between the proportions of programs offered between MSI and non-MSI schools.

```
        Design-based t-test

data:  offers_stats ~ Strata
t = -1.6114, df = 267, p-value = 0.1083
alternative hypothesis: true difference in
mean is not equal to 0
95 percent confidence interval:
 -0.21469788  0.02144036
sample estimates:
difference in mean
       -0.09662876
```

```
        Design-based t-test

data:  offers_ds ~ Strata
t = -0.17576, df = 267, p-value = 0.8606
alternative hypothesis: true difference in
mean is not equal to 0
95 percent confidence interval:
 -0.1371214  0.1146465
sample estimates:
difference in mean
       -0.01123747
```

Based on an alpha of 0.05, we fail to reject both null hypotheses, and can conclude that there is no significant difference in the proportion of statistics and data science majors/minors between MSI and non-MSI schools. This is further supported by zero being captured by both confidence intervals, meaning that the true difference in proportion for the population may be zero.

1.3. Do selective schools have higher rates of data science major compared to inclusive schools?

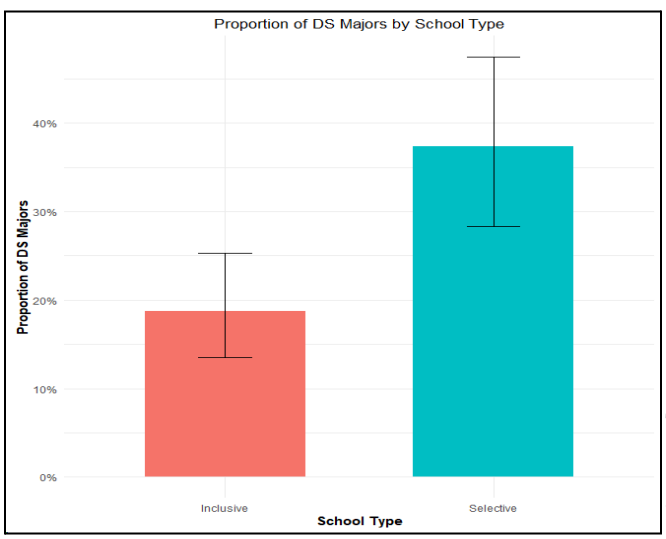
To answer this question, we first selected the relevant columns, Strata and DS Major, and created a binary grouping based on school type (Selective vs. Inclusive) to compare the rates of data science majors between these two groups.

Similar to previous questions, in this part we also used the survey package. The analysis was done in three main steps, first the proportion estimation, which we calculated the percentage of "Yes" for data science majors in each strata, next, t-test in which we compared the proportion using svyttest() in R, and at last, we found the 95% confidence intervals for data science major proportions in each stratum.

We observe the following:

	Inclusive School	Selective School
Proportion Estimates	18.7%	37.4%
SE	2.99%	4.89%
CI	(13.5%, 25.3%)	(28.3%, 47.5%)

The result of t-test suggests the t-value of 3.26 which is relatively large and suggests there is a significant difference between the two proportions, degrees of freedom of 267, and p-value of 0.00127 which is highly significant and matches our previous conclusion.



* This figure shows the proportion of DS majors in each school with their confidence intervals.

The proportion difference between two schools is 18.7% with Confidence interval falling between (7.38%, 29.94%).

Considering all the findings above, we can reject the Null hypothesis and conclude that selective schools have a significantly higher proportion of DS majors, compared to inclusive schools.

1.4. Do selective schools offer both major and minor in statistics more frequently compared to inclusive schools?

Statistics can usually be found in universities more often as a minor over a major. However, as a growing field, universities incorporating a Statistics major is becoming more and more frequent. To answer this question, we used R to separate the schools that were selective (S) or inclusive (I). Using an if/then statement, we assigned a binary value to whether or not the school had both a major and a minor. For example, if both the variables Stat.major and Stat.Minor were yes, then the value would be 1. If one or both of those variables contained a no, then the value was 0.

Once those variables were assigned, we used the svymean() function in R to find the proportion of schools that have both a Statistics major and minor between different levels of selectivity. The following table shows the given values:

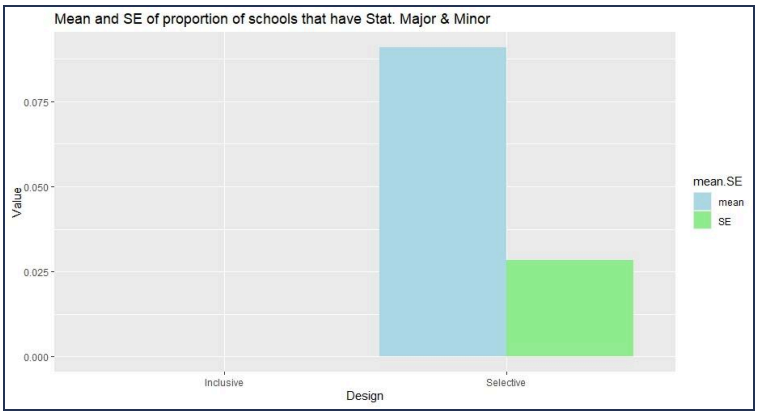
Design	**Mean**	**SE**
Selective	0.09091	0.02843
Inclusive	0	0

While at first these results were a bit confusing, taking into account that the Inclusive schools often included Community Colleges or other non-four-year universities, these values made sense. To see whether or not these values had a significant difference, we ran a t-test.

```
        Design-based t-test

data:  Stat_Major_Minor ~ Design
t = 3.2, df = 171, p-value = 0.002
alternative hypothesis: true difference in mean is not equal to 0
95 percent confidence interval:
 0.03479 0.14703
sample estimates:
difference in mean
           0.09091
```

Even though this may have been a given, the p-value of 0.002 indicates that there is a significant difference between the selective schools that have both a Statistics major and minor, and the inclusive schools which have the like. Because we didn't have values for the inclusive schools, we got a confidence interval for the Selective schools. With 95% confidence, we can say that the proportion of selective schools that have both a major and minor is captured within the interval (0.03518, 0.1466). Below is a visualization of these findings.



2. Course Level

2.1. Is there a difference in the proportion of schools that require math prerequisites for statistics degrees and data science degrees?

At first with this question, we were going to go with a Calculus prerequisite. However, after looking through the data, we found that a Mathematics prerequisite might be better to look at because many schools did not have a calculus prerequisite. We focused on Data Science and Statistics classes, as those majors were what the rest of the project was studying as well. We separated the data out by the Department and whether or not there was a Math prerequisite for the class. Like with the final University level question, we assigned binary values to results. If the class was marked as yes for the math prereq., it was assigned a 1. If not, it was assigned a no.

Figuring out the best way to find these proportions was a bit difficult at first, but we ended up on using the svyby() function to properly gain these values. After using it in R, we got these results:

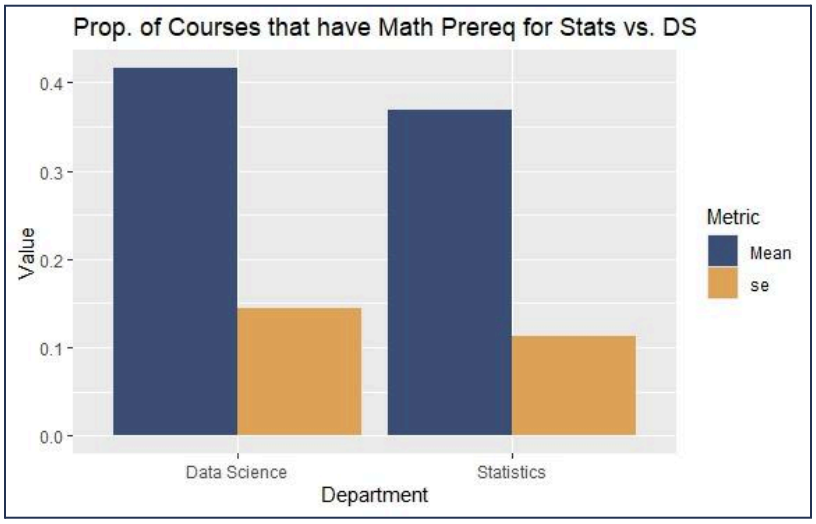
Department	**Mean**	**SE**	**Lower_CI**	**Upper_CI**
Data Science	0.4167	0.1447	0.1331	0.7002
Statistics	0.3684	0.1125	0.1479	0.5889

Just looking by this output, there doesn't seem to be a very large difference between the two mean & standard error values or the confidence intervals. We can see that Data Science has more courses with Math Prerequisites from this table, but Statistics is not that far behind. So, we ran a t-test for the significance.

```
        Design-based t-test

data:  Math_Pre_Binary ~ Department
t = -0.26, df = 29, p-value = 0.8
alternative hypothesis: true difference in mean is not equal to 0
95 percent confidence interval:
 -0.4231  0.3266
sample estimates:
difference in mean
          -0.04825
```

As we can see from this t-test, the t value sits at -0.26, with the p-value being 0.8. This is indicative that there is not a strong difference between the proportion of Data Science classes that have a Math prerequisite and the proportion of Statistics classes that have a Math prerequisite. This is also very evident in the visualization.



Discussion

In our study, we explored the characteristics of statistics and data science programs across selected US colleges and universities. Using the MASDER dataset, we were able to analyze and compare trends to answer our research questions. We believe that as the fields of statistics and data science continue to grow, more incoming students will be interested in entering these programs. Our research questions aimed to answer big questions that SDS students might have when thinking about what kind of institution they want to attend. Our results reveal some interesting insights of interest:

1. We found no significant difference in the proportion of statistics majors offered between inclusive and selective Bach/Grad institutions. We can also note that none of the colleges offering Associate's degrees also offer a statistics major. This is likely because the statistics career field often requires a Bachelor’s degree at entry level, and the field is still relatively small.

2. We concluded there was no significant difference in the proportion of SDS majors or minors between MSI and non-MSI schools. Depending on the alpha level, it could be argued there is a significant difference of statistics majors/minor between MSI and non-MSI schools, but further analysis would be required to make a final decision.
3. We found that selective schools have a significantly higher proportion of data science majors compared to inclusive schools. This cound be because of a higher demand, better access to resources, or a school's flexibility that allows them to offer a data science major.
4. For this question, we found that no inclusive schools in our sample offered both a statistics major and minor, whereas about 9.1% of selective schools did. This is likely for the same reasons as our results in the previous question.
5. We concluded there was no significant difference in the proportion of statistics and data science courses that require a math prerequisite. This may be because there is some overlap between the two degrees, so the math prerequisites are required at a similar rate.

Our findings show key trends in the distribution of statistics and data science programs in the U.S. institutions. Limitations such as sample size constraints and potential gaps in the data may affect the precision and scope of our analysis. Future research could explore longitudinal trends, curriculum content, and student outcomes to better understand how these programs evolve and impact workforce readiness.

When looking at our results overall, we can notice that our own school, University of Nebraska-Lincoln, is an inclusive school that offers both a statistics major and minor. This is a recent development, and the statistics major has only existed for the past few years. Depending on how long ago the MASDER data was collected, it may no longer accurately represent this rapidly growing field as more and more institutions start offering SDS programs.

There are also many factors not included in the MASDER dataset that would likely be of interest in a further study. Such factors include the size of the school, level of funding, or post-introductory SDS courses offered. This research would allow a new avenue of analysis into more course level understanding. Continuing research into this topic and watching the trend of SDS programs over time would not only deepen our understanding, but also give insight on the growing labor market for these two fields.

Finally, expanding the scope of educational research to consider more factors would be extremely valuable for understanding equity in education. By examining how SDS programs are represented in our different strata, we can identify strategies to create a more inclusive and effective pipeline into the field. Overall, this type of research and analysis helps institutions design and refine programs that ensure all students have a chance to succeed and build opportunities for the next generation of statisticians and data scientists.

Citations

Encyclopædia Britannica, inc. (2024, December 6). *Statistics*. Encyclopædia Britannica. https://www.britannica.com/science/statistics

U.S. Bureau of Labor Statistics. (2024, August 29). *Fastest growing occupations*. U.S. Bureau of Labor Statistics. https://www.bls.gov/emp/tables/fastest-growing-occupations.htm

Code Appendix

<u>Data Cleaning & Merge:</u>

```
#merge University of COurse datasets
getwd()
university <- read.csv("masder-university.csv")
course <- read.csv("masder-course.csv")

#clean the data--> missing and duplicated in UNITID col.
university <- university[!is.na(university$UNITID), ]
course <- course[!is.na(course$UNITID), ]

#merge datasets --> by UNITID column
merged_data <- merge(university, course, by="UNITID", all=TRUE)

write.csv(merged_data, "merged.csv", row.names = FALSE)

masder <- read.csv("merged.csv")
head(masder)

#missing data
colSums(is.na(masder))

#check duplicates
masder <- masder[!duplicated(masder$UNITID), ] #!duplicated(merged_data[c("UNITID",
"Course.Name")] ?? if needed
                                        ########################################
```

<u>Question 1:</u>

```
library(survey)
library(dplyr)
library(tidyverse)
```

```
library(ggplot2)

#Select only necessary columns and transform the data into binary#
uni1 <- uni %>%
  select("Strata","Intro.Stat.Course","Stat.major","UNITID") %>%
  mutate(statmajorprop = ifelse(Stat.major == "Yes", 1, 0)) %>%
  drop_na(UNITID)

#create the strata dataframe#
strata_info <- data.frame(
  Strata = c("GSN", "GIN", "*SM", "*IM", "BSN", "BIN", "AxM", "AxN"),
  N_h = c(438, 386, 64, 207, 232, 315, 259, 850),
  n_h = c(19, 18, 12, 12, 11, 16, 11, 48)
)

#create new column with sample weight, then mutate the dataframes#
strata_info <- strata_info %>%
  mutate(sampwt = N_h / n_h)
uni1 <- uni1 %>%
  left_join(strata_info, by = "Strata")

###ACTUAL DESIGN AND ANALYSIS###
unides <- svydesign(id=~1, strata=~Strata, weights=~sampwt, fpc=~N_h,data=uni1)
strmean <- tapply(uni1$statmajorprop,uni1$Strata, mean)
strmean
strvar <- tapply(uni1$statmajorprop,uni1$Strata, var)
strvar
svyby(~statmajorprop, by=~Strata,unides,svymean, keep.var=TRUE)
svytotal(~statmajorprop, unides)

confint(svytotal(~statmajorprop, unides))


###graph of outputs###
svyby(~statmajorprop, by=~Strata,unides,svymean, keep.var=TRUE)

strata_summary <- data.frame(
  Strata = c("*IM", "*SM", "AxM", "AxN",
             "BIN", "BSN", "GIN", "GSN"),
  Mean = c(0.0833, 0.3333, 0, 0, 0.0625, 0.0909, 0, 0.3158),
```

```
  SE = c(0.08088, 0.12812, 0, 0, 0.06089, 0.08873, 0, 0.10716)
)

strata_summary <- strata_summary %>%
  mutate(LowerCI = Mean - 1.96 * SE,
         UpperCI = Mean + 1.96 * SE)

ggplot(strata_summary, aes(x = Strata, y = Mean)) +
  geom_point(size = 3, color="red") +
  geom_errorbar(aes(ymin = LowerCI, ymax = UpperCI), width = 0.2, color="blue") +
  labs(
    title = "Strata Mean Proportions with Confidence Intervals",
    x = "Strata",
    y = "Mean Proportions"
  )

####t-test to compare *IM and *SM###
# Inputs for the two strata
mean1 <- 0.08333
se1 <- 0.08088
n1 <- 207

mean2 <- 0.03333
se2 <- 0.12812
n2 <- 64

# Pooled SE
pooled_se <- sqrt(se1^2 + se2^2)

t_stat <- (mean1 - mean2) / pooled_se

# Welch df
df <- (se1^2 + se2^2)^2 /
  ((se1^4 / (n1 - 1)) + (se2^4 / (n2 - 1)))

# Two-tailed p-value
p_value <- 2 * pt(-abs(t_stat), df)

# Output results
list(t_stat = t_stat, df = df, p_value = p_value)
```

##

Question 2:

```
data <- read.csv("merged.csv")
library(dplyr)
library(stringr)
library(survey)
library(tidyr)
library(ggplot2)

############POPULATION LEVEL

####all msi schools
msi <- data %>%
  filter((str_detect(Strata, "M"))) %>%
  mutate(Strata = ifelse(grepl("M", Strata), "M", Strata))

#create binary columns for whether or not they offer programs
msi_mm <- msi %>%
  select("Strata", "Stat.major", "Stat.Minor", "DS.Major", "DS.Minor",
"School..official.from.college.scorecard..x") %>%
  mutate(
    offers_stats_msi = ifelse(Stat.major == "Yes" | Stat.Minor == "Yes", 1, 0),
    offers_ds_msi = ifelse(DS.Major == "Yes" | DS.Minor == "Yes", 1, 0))

design_msi <- svydesign(ids = ~1, strata = ~Strata, weights = ~1, data = msi_mm)

#proportion of MSI schools offering statistics programs
prop_stat_msi <- svymean(~offers_stats_msi, subset(design_msi, Strata == Strata)) #0.299
prop_stat_msi
confint(prop_stat_msi)

prop_ds_msi <- svymean(~offers_ds_msi, subset(design_msi, Strata == Strata)) #0.338
prop_ds_msi
confint(prop_ds_msi)


####all non-msi schools
non_msi <- data %>%
  filter((str_detect(Strata, "N"))) %>%
  mutate(Strata = ifelse(grepl("N", Strata), "N", Strata))
```

```
#create binary columns
non_msi_mm <- non_msi %>%
  select("Strata", "Stat.major", "Stat.Minor", "DS.Major", "DS.Minor",
"School..official.from.college.scorecard..x") %>%
  mutate(
    offers_stats_nonmsi = ifelse(Stat.major == "Yes" | Stat.Minor == "Yes", 1, 0),
    offers_ds_nonmsi = ifelse(DS.Major == "Yes" | DS.Minor == "Yes", 1, 0))

design_non <- svydesign(ids = ~1, strata = ~Strata, weights = ~1, data = non_msi_mm)

# proportion of non-MSI schools offering statistics programs
non_msi_stats <- svymean(~offers_stats_nonmsi, subset(design_non, Strata == Strata)) #0.202
non_msi_stats
confint(non_msi_stats)

non_msi_ds <- svymean(~offers_ds_nonmsi, subset(design_non, Strata == Strata)) #0.326
non_msi_ds
confint(non_msi_ds)

#############STRATUM LEVEL
#combine msi_mm and non_msi_mm to make one design
combined <- bind_rows(
  msi_mm %>% mutate(Group = "MSI"),
  non_msi_mm %>% mutate(Group = "Non-MSI"))

design <- svydesign(ids = ~1, strata = ~Strata, weights = ~1, data = combined)

# Statistics programs for each stratum
svyby(~offers_stats, ~Strata, design, svymean)

# Data science programs for each stratum
svyby(~offers_ds, ~Strata, design, svymean)

# running t-test on difference between MSI and NON-MSI schools
svyttest(offers_stats ~ Strata, design)

svyttest(offers_ds ~ Strata, design)

#visualization
```

```
combinedv <- combined %>%
  pivot_longer(cols = c(offers_stats, offers_ds), names_to = "Program", values_to = "Offered")
%>%
  group_by(Strata, Program) %>%
  summarise(Proportion = mean(Offered),
            SE = sd(Offered) / sqrt(n())) %>%
  mutate(Strata = recode(Strata, offers_stats = "Stats", offers_ds = "Data Science"),
         Program_School = paste(Program, Strata, sep = " - "))

ggplot(data = combinedv, aes(x = Program_School, y = Proportion)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.7), width = 0.5) +
  geom_errorbar(aes(ymin = Proportion - SE, ymax = Proportion + SE),
                position = position_dodge2(width = 0.7, padding = 0.2), width = 0.25) +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Prop of Stats and Data Science Majors or Minors",
    x = "Strata",
    y = "Proportion",
    fill = "School Type") +
  scale_x_discrete(labels = c(
    "offers_stats - M" = "Stats - MSI",
    "offers_ds - M" = "Data Science - MSI",
    "offers_stats - N" = "Stats - Non-MSI",
    "offers_ds - N" = "Data Science - Non-MSI")) +
  theme_minimal() +
  geom_text(aes(label = scales::percent(Proportion, accuracy = 0.1)),
            vjust = 8, size = 3, color = "white")
```

##################################

<u>Question 3:</u>

```
> library(survey)
> library(dplyr)
> library(ggplot2)
> masder <- read.csv("merged.csv")
>
> DS <- masder %>%
+   select(Strata, DS.Major, DS.Minor)
> #1,0 <- yes, no for ds minor and major
> DS$DS.Major <- factor(DS$DS.Major, levels = c("No", "Yes"))
```

```
> DS$DS.Minor <- factor(DS$DS.Minor, levels = c("No", "Yes"))
>
> Selective <- DS  %>% filter(grepl("S", Strata))
> Inclusive <- DS %>% filter(grepl("I", Strata))
>
> #survey design
> design <- svydesign(ids = ~1, strata = ~Strata, data = DS, weights = ~1)
>
> # Estimate proportions of "Yes" for DS.Major
> prop_estimates <- svyby(~DS.Major, ~Strata, design, svymean, na.rm = TRUE)
>
> DS$Strata_final <- ifelse(grepl("S", DS$Strata), "Selective", "Inclusive")
>
> # t-test
> svy_ttest <- svyttest(DS.Major == "Yes" ~ Strata_final, design_binary)
> print(svy_ttest)

        Design-based t-test

data:  DS.Major == "Yes" ~ Strata_final
t = 3.2567, df = 267, p-value = 0.001273
alternative hypothesis: true difference in mean is not equal to 0
95 percent confidence interval:
 0.07378814 0.29941760
sample estimates:
difference in mean
         0.1866029

>
> design <- svydesign(ids = ~1, strata = ~Strata_final, data = DS, weights = ~1)
>
> # CI for S
> ci_selective <- svyciprop(~I(DS.Major == "Yes"), subset(design, Strata_final == "Selective"))
> print(ci_selective)
                        2.5% 97.5%
I(DS.Major == "Yes") 0.374 0.283 0.475
>
> # CI for I
> ci_inclusive <- svyciprop(~I(DS.Major == "Yes"), subset(design, Strata_final == "Inclusive"))
> print(ci_inclusive)
```

```
                    2.5% 97.5%
I(DS.Major == "Yes") 0.187 0.135 0.253
```

##

Question 4:

```
#Plug in Nh values
selective_Nh <- c(438, 64, 232, 850)
inclusive_Nh <- c(387, 207, 315, 259)

#Seperate out Strata
selective_ds <- merged_data  %>% filter(grepl("S", Strata))
inclusive_ds <- merged_data %>% filter(grepl("I", Strata))

#Use for 'manual' fpc calculation
selective_ds$Nh <- rep(selective_Nh, each = length(selective_ds) / length(selective_Nh))
inclusive_ds$Nh <- rep(inclusive_Nh, each = length(inclusive_ds) / length(inclusive_Nh))

#assign binary values to get proportions
merged_data$Stat_Major_Minor <- ifelse(merged_data$Stat.major == 'Yes' &
merged_data$Stat.Minor == 'Yes', 1, 0)

selective_design <- svydesign(ids = ~1, strata = ~Strata, data = selective_ds, weights = NULL,
fpc = ~Nh)
inclusive_design <- svydesign(ids = ~1, strata = ~Strata, data = inclusive_ds, weights = NULL,
fpc ~Nh)

selective_stat_mm <- svymean(~Stat_Major_Minor, design = selective_design, na.rm = TRUE)
inclusive_stat_mm <- svymean(~Stat_Major_Minor, design = inclusive_design, na.rm = TRUE)

selective_stat_mm
inclusive_stat_mm

#Create data frame and bind for visualization purposes
selective_df <- as.data.frame(selective_stat_mm)
inclusive_df <- as.data.frame(inclusive_stat_mm)

df_merged <- rbind(
  cbind(Design = 'Selective', as.data.frame(selective_stat_mm)),
  cbind(Design = 'Inclusive', as.data.frame(inclusive_stat_mm))
)
```

```
df_merged <- df_merged %>%
  rename(
    SE = Stat_Major_Minor
  )

df_long <- df_merged %>%
  gather(key = mean.SE, value = Value, mean:SE)

ggplot(df_long, aes(Design, Value, fill = mean.SE)) + geom_col(position = 'dodge') +
scale_fill_manual(values = c('lightblue', 'lightgreen')) + labs(title = 'Mean and SE of proportion
of schools that have Stat. Major & Minor')

#95% Confidence Interval
confint(selective_stat_mm)

selective_ds$Design <- "Selective"
inclusive_ds$Design <- "Inclusive"

combined_ds <- rbind(selective_ds, inclusive_ds)
combined_design <- svydesign(ids = ~1, strata = ~Strata, data = combined_ds, weights = NULL)

#t-test for significance
ttest_result <- svyttest(Stat_Major_Minor ~ Design, design = combined_design)
ttest_result
```

<u>Question 5:</u>

```
#Select only the columns that are needed and find the totals
prereq <- merged_data %>%
  select(Department, Math.Prereq) %>%
  na.omit() %>%
  group_by(Department) %>%
  filter(Department == 'Statistics' | Department == "Data Science") %>%
  group_by(Department) %>%
  summarize(Nh = n())

prereq$Math_Pre_Binary <- ifelse(prereq$Math.Prereq == 'Yes', 1, 0)

prereq_design <- svydesign(ids = ~1, data = prereq)
```

```
prop_math_prereq <- svymean(~Math_Pre_Binary,strata = ~Department, design =
prereq_design, fpc = ~Nh)
prop_math_prereq

prop_dept <- svyby(~Math_Pre_Binary, ~Department, design = prereq_design, FUN = svymean,
na.rm = TRUE)
prop_dept

prop_dept_df <- as.data.frame(prop_dept)

prop_dept_df <- prop_dept_df %>%
  rename(
    Mean = Math_Pre_Binary
  ) %>%
  select(Department, Mean, se) %>%
  gather(key = 'Metric', value = 'Value', Mean, se)

ggplot(prop_dept_df, aes(x = Department, y = Value, fill = Metric, position = 'dodge')) +
  geom_bar(stat = 'identity', position = 'dodge') +
  labs(
    title = "Prop. of Courses that have Math Prereq for Stats vs. DS"
  ) +
  scale_fill_manual(values = c('#3C4F76', '#E0A458'))

svyttest(Math_Pre_Binary ~ Department, prereq_design)

prop_dept_ci <- svyby(~Math_Pre_Binary, ~Department, design = prereq_design, FUN =
svymean, na.rm = TRUE, vartype = c('se', 'ci'))
prop_dept_ci

prop_dept_ci_df <- prop_dept_ci %>%
  rename(
    Mean = Math_Pre_Binary,
    Lower_CI = ci_l,
    Upper_CI = ci_u
  )

prop_dept_ci_df
```